ABSTRACT

While Large Language Models (LLMs) like GPT and LLaMA have revolutionised how we access information, they suffer from a rather significant blind spot: they are static. Once trained, these models are essentially frozen in time, often providing answers that were true years ago but are now factually obsolete. This problem is known as "temporal drift." This project addresses that limitation by developing T-RAG, an integrated framework that helps language models understand not just what a fact is, but when it is valid.

Instead of relying on standard retrieval methods that only look for similar keywords, our system introduces the dimension of time in its Retrieval-Augmented Generation (RAG) approach. We developed a Temporal Knowledge Graph (TKG) using Neo4j, which stores data as events with timestamps rather than flat text, allowing the system to track how relationships between entities change over time. To ensure the model doesn't use expired information, we developed an Adaptive Deprecation Detection (ADD) module. This uses a machine learning-based "half-life" decay model to calculate a freshness score for every fact, automatically filtering out data that has become obsolete.

The core of our solution consists of a hybrid retrieval engine that ranks the timestamped information using a Weighted Relevance Score (WRS). This combines the semantic meaning of a user's query with the temporal validity, i.e., the validity of the available data with respect to a given time period, ensuring the LLM is fed only the most current context. The framework is built using datasets like ICEWS14 and Wikidata Temporal, and its performance is measured with metrics such as Mean Reciprocal Rank (MRR) and Temporal Accuracy. The results demonstrate that T-RAG enhances accuracy in the final output and reduces response times, making it a viable solution for time-sensitive applications like news verification and financial forecasting.

Keywords: Large Language Models, Temporal Drift, Retrieval-Augmented Generation, Temporal Knowledge Graphs, Adaptive Deprecation Detection.

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LIST OF ABBREVIATIONS

Abbreviation	Full Description
LLMs	Large Language Models
T-RAG	Time-Aware Retrieval-Augmented Generation
TKG	Temporal Knowledge Graphs
ADD	Adaptive Deprecation Detection
MRR	Mean Reciprocal Rank
TA	Temporal Accuracy
EYFS	Early Years Foundation Stage
UBA	User Behavior Analytics
LLM-PD	Large Language Model Empowered Proactive Defense
XAI	Explainable Artificial Intelligence
FVS	Fact Validity Score
WRS	Weighted Relevance Score
GPT	Generative Pre-trained Transformer
LLaMA	Large Language Model Meta AI
TCS	Temporal Consistency Score
RE-GCN	Relational Graph Convolutional Network
LLM-DA	LLM-Guided Dynamic Adaptation
NER	Named Entity Recognition

GRU	Gated Recurrent Unit
ACL	Association for Computational Linguistics
NeurIPS	Neural Information Processing Systems
EMNLP	Empirical Methods in Natural Language Processing
LREC	Language Resources and Evaluation
GPU	Graphics Processing Unit
RAM	Random Access Memory
SSD	Solid State Drive
NVMe	Non-Volatile Memory Express
OS	Operating System
API	Application Programming Interface
RSS	Rich Site Summary (or Really Simple Syndication)
JSON	JavaScript Object Notation
BERT	Bidirectional Encoder Representations from Transformers
FAISS	Facebook AI Similarity Search
DB	Database

INTRODUCTION

1.1 Overview

Large Language Models like GPT and LLaMA have achieved unprecedented results across reasoning and knowledge-intensive tasks. However, their static knowledge base introduces a crucial limitation: temporal drift, whereby models produce outdated or contextually invalid information. Very recent works on temporal knowledge representation and retrieval-augmented generation have demonstrated that dynamic external knowledge can indeed alleviate this problem. Temporal Retrieval-Augmented Generation, or T-RAG, seeks to blend the advantages of Temporal Knowledge Graphs with time-aware retrieval in order to guarantee updated, explainable, factually accurate results.

1.2 Problem Statement

And in this context, one vital shortcoming of existing LLMs is that their knowledge becomes stale; they do not recognize when the facts they learned go invalid. Similarly, traditional RAG systems retrieve documents that are semantically relevant to a query but ignore temporal validity cues, which often leads to misinformation.

Therefore, there is an urgent requirement of the temporal reasoning mechanism that can:

(i) find obsolete facts

(ii) retrieve time-aligned knowledge

(iii) keep generative outputs consistent with evolving real-world facts.

1.3 Objectives

- To design a Temporal Knowledge Graph for structured, time-stamped fact representation.
- To develop an Adaptive Deprecation Detection model that identifies outdated knowledge.

- There is the implementation of the T-RAG framework, incorporating temporal reasoning.
- To evaluate system performance on temporal datasets such as ICEWS14 and Wikidata Temporal using metrics like MRR, Hits@k, and Temporal Accuracy (TA).
- To ensure explainability through timestamped evidence and provenance tracking.

1.4 Motivation

The knowledge of the world keeps evolving. For example, the political leadership, scientific knowledge, and rules and regulations change with time. LLMs trained on fixed data are unable to adapt to such evolution. Inspired by recent advances in temporal knowledge reasoning [2], [6] and dynamic LLM adaptation [8], the work in this paper targets constructing an intelligent system that learns not only what to answer but also when the answer was valid.

LITERATURE SURVEY

Sl. No.	Paper Title	Authors	Core Contribution	Techniques	Relevance to the Proposed Work
1	*Outdated Fact Detection in Knowledge Bases*	Hao et al.	Introduces a method to identify outdated facts in static knowledge bases.	Fact validity prediction using temporal decay and rule-based heuristics.	Forms the **foundation of deprecation detection**; inspires the Adaptive Deprecation Detection (ADD) module.
2	*HALO: Half-Life-Based Outdated Fact Filtering in Temporal Knowledge Graphs*	Ding et al.	Proposes a half-life model to dynamically measure fact relevance in temporal KGs.	Time decay functions with exponential half-life weighting.	Directly influences the **half-life scoring** used in the ADD module to filter obsolete information.
3	*Chain-of-History Reasoning for Temporal Knowledge Graph Forecasting*	Xia et al.	Develops multi-step reasoning chains for time-evolving graphs.	Chain-of-History (CoH) reasoning framework with LLM integration.	Provides the conceptual base for **temporal reasoning** and **multi-hop inference** in the Time-Aware RAG pipeline.
4	*LLM-Guided Dynamic Adaptation for Temporal Knowledge Graph Reasoning (LLM-DA)*	Wang et al.	Introduces an LLM-based mechanism to update temporal rules dynamically.	LLM-driven rule generation and self-adaptive reasoning.	Inspires the **dynamic adaptation** logic for updating time-sensitive retrieval patterns in your T-RAG.
5	*Is Your LLM Outdated? Evaluating LLMs at Temporal Generalization*	Karpas et al.	Evaluates LLMs' performance degradation over time due to static training.	FreshBench benchmark for temporal generalization testing.	Motivates the **temporal evaluation metrics** (e.g., time-specific accuracy) used in your system testing.
6	*History Matters: Temporal Knowledge Editing in LLMs*	Yin et al.	Proposes a temporal knowledge editing framework to preserve historical facts while updating new ones.	Multi-Editing with Time Objective (METO) and ATOKE benchmark.	Reinforces your project's goal to **update without forgetting** — maintaining both current and historical knowledge.
7	*Outdated Facts Make Language Models Underperform in Temporal Reasoning*	Xu et al.	Demonstrates that outdated facts severely degrade LLM reasoning accuracy.	TEMPURA benchmark for temporal reasoning.	Strengthens the motivation for **temporal alignment** in LLM-based RAG systems.

8	*Test of Time: A Benchmark for Evaluating LLMs on Temporal Understanding*	Huang et al.	Provides a standardized benchmark for testing LLMs' time consistency.	Temporal Consistency Score (TCS) and Relative Temporal Error (RTE).	Informs the **evaluation methodology** for assessing the time consistency of generated responses.
9	*Dynamic Hypergraph Embedding for Temporal Knowledge Graph Reasoning (DHE-TKG)*	Liu et al.	Captures high-order temporal dependencies using hypergraph embeddings.	Dynamic Hypergraph Networks and temporal fusion.	Inspires your **Temporal Knowledge Graph (TKG)** construction with **multi-entity temporal aggregation**.
10	*A Survey on Temporal Knowledge Graphs: Representation Learning and Applications*	Cai et al.	Provides a comprehensive taxonomy of TKG modeling approaches.	Categorization of 10 families: translation, GNN, autoregressive, TPP, etc.	Establishes the **theoretical base** for your system's temporal graph representation and reasoning model.
11	*Retrieval-Augmented Generation: A Comprehensive Survey of Architectures, Enhancements, and Robustness Frontiers*	Sharma et al.	Summarizes developments in RAG architectures and robust retrieval.	Retriever-centric, generator-centric, and hybrid RAG frameworks.	Provides the structural backbone for your **Time-Aware RAG (T-RAG)** pipeline.
12	*FreshBench: A Dynamic Benchmark for Temporal Generalization*	Zhu et al.	Introduces FreshBench to assess temporal biases and degeneration in LLMs.	Bits-per-character (BPC) compression metric, forecasting accuracy.	Used to benchmark **temporal generalization** and validate that your model resists **nostalgia bias** (outdated preference).
13	*Temporal Knowledge Graph Reasoning with Dynamic Hypergraph Embedding (DHE-TKG)*	Liu et al.	Models higher-order interactions in evolving KGs.	GRU + hypergraph embedding fusion.	Reinforces your **temporal graph encoding** and event evolution modeling.
14	*TimeBench: Benchmarking Time-Aware Reasoning in LLMs*	Kim et al.	Evaluates how LLMs process and reason about explicit time cues.	Question answering with temporal tagging and contextual grounding.	Helps design **time-conditioned prompting** in your LLM generation layer.

REQUIREMENT ANALYSIS & DESIGN

3.1 Analysis

- Problem: Static knowledge in LLMs leads to outdated outputs.
- Solution: Introduce a hybrid system that combines semantic retrieval with temporal validity scoring.
- Modules: TKG, ADD, Time-Aware Retriever, LLM Generator, and Verifier.
- Constraints: Scalability, low latency (<3s), explainability.

3.2 Hardware Requirements

Component	Minimum	Recommended
Processor	Intel i5/ AMD Ryzen 5	Intel i7/ AMD Ryzen 7
RAM	16 GB	32 GB
GPU	GTX 1050Ti (3 GB)	RTX 3070 (8 GB)
Storage	256 GB SSD	1 TB NVMe
Network	100 Mbps	1 Gbps

3.3 Software Requirements

- Language: Python 3.10+
- Frameworks: FastAPI, PyTorch, Neo4j, FAISS, pandas
- Databases: PostgreSQL, Neo4j, Milvus
- Libraries: Transformers, Sentence-BERT, scikit-learn
- OS: Ubuntu 22.04 / Windows 11

3.4 System Architecture Diagram

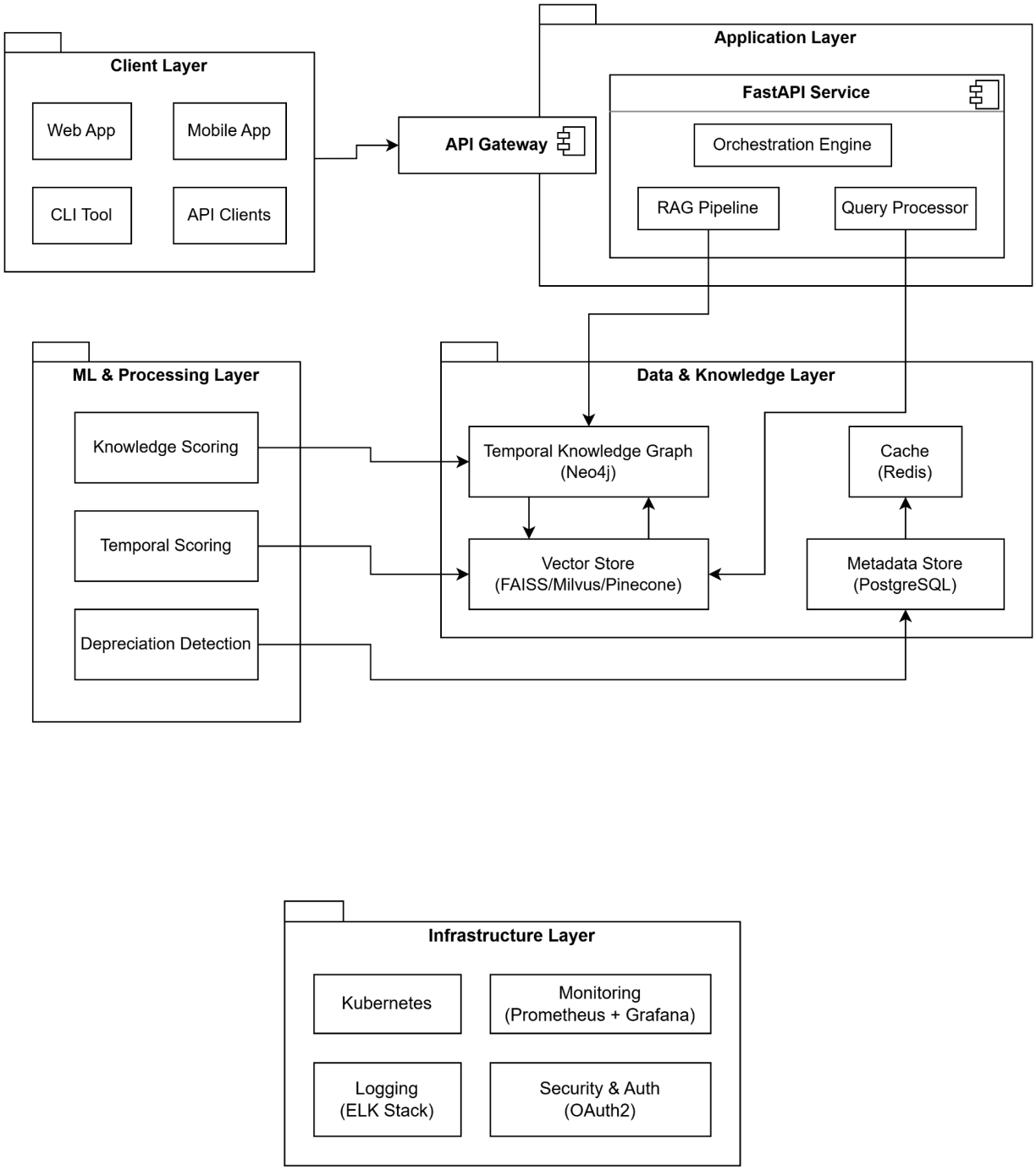

Fig. 1 - System Architecture Diagram

3.5 Data Flow Diagram

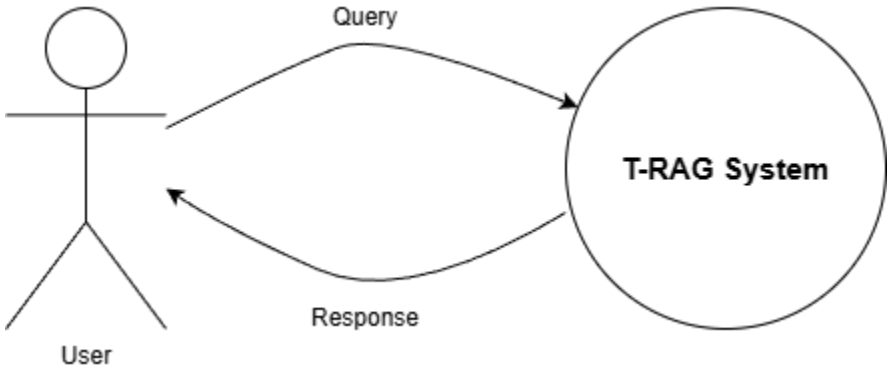

Fig. 2 - Level 0 Data Flow Diagram

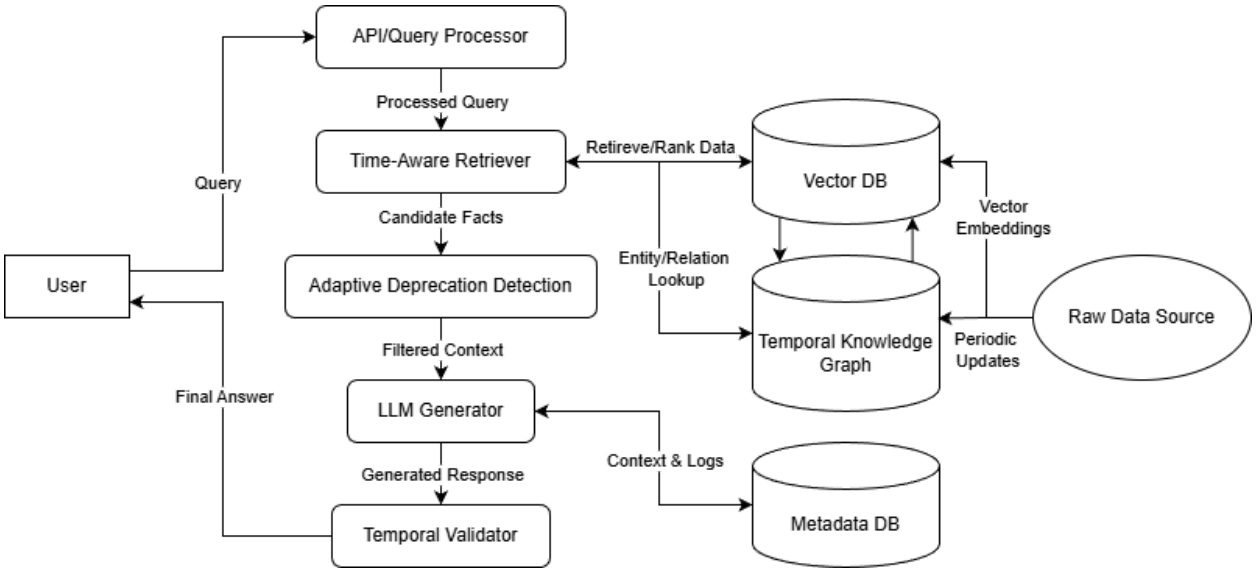

Fig. 3 - Level 1 Data Flow Diagram

3.6 Use Case Diagram

Fig. 4 - Use Case Diagram

3.7 Activity Diagram

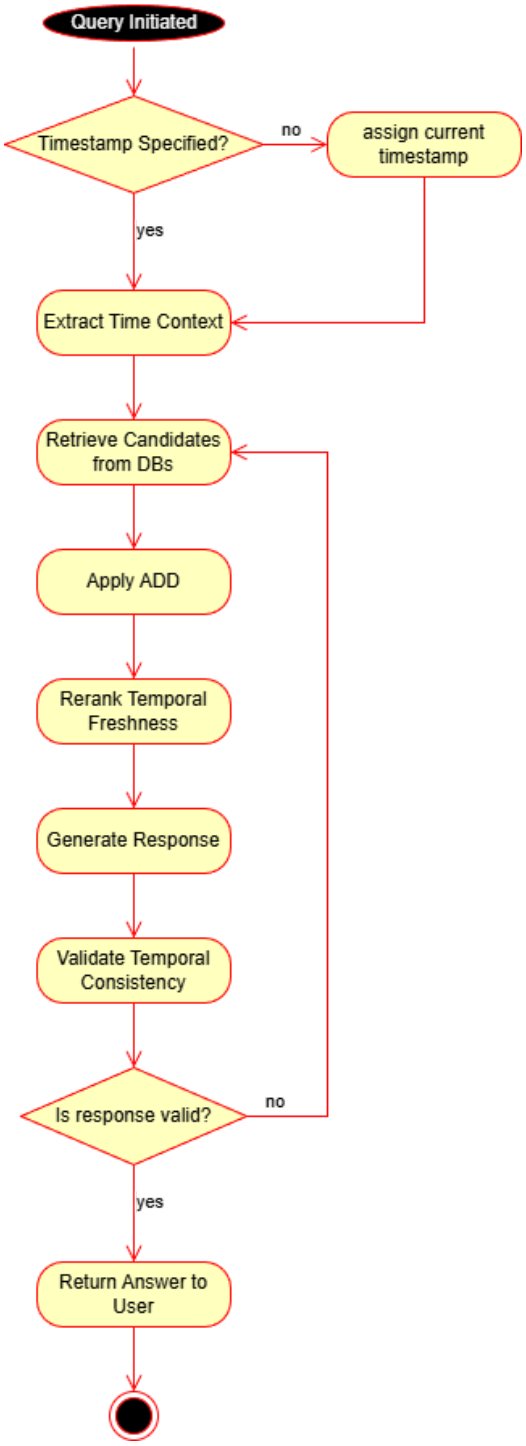

Fig. 5 - Activity Diagram

3.8 Sequence Diagram

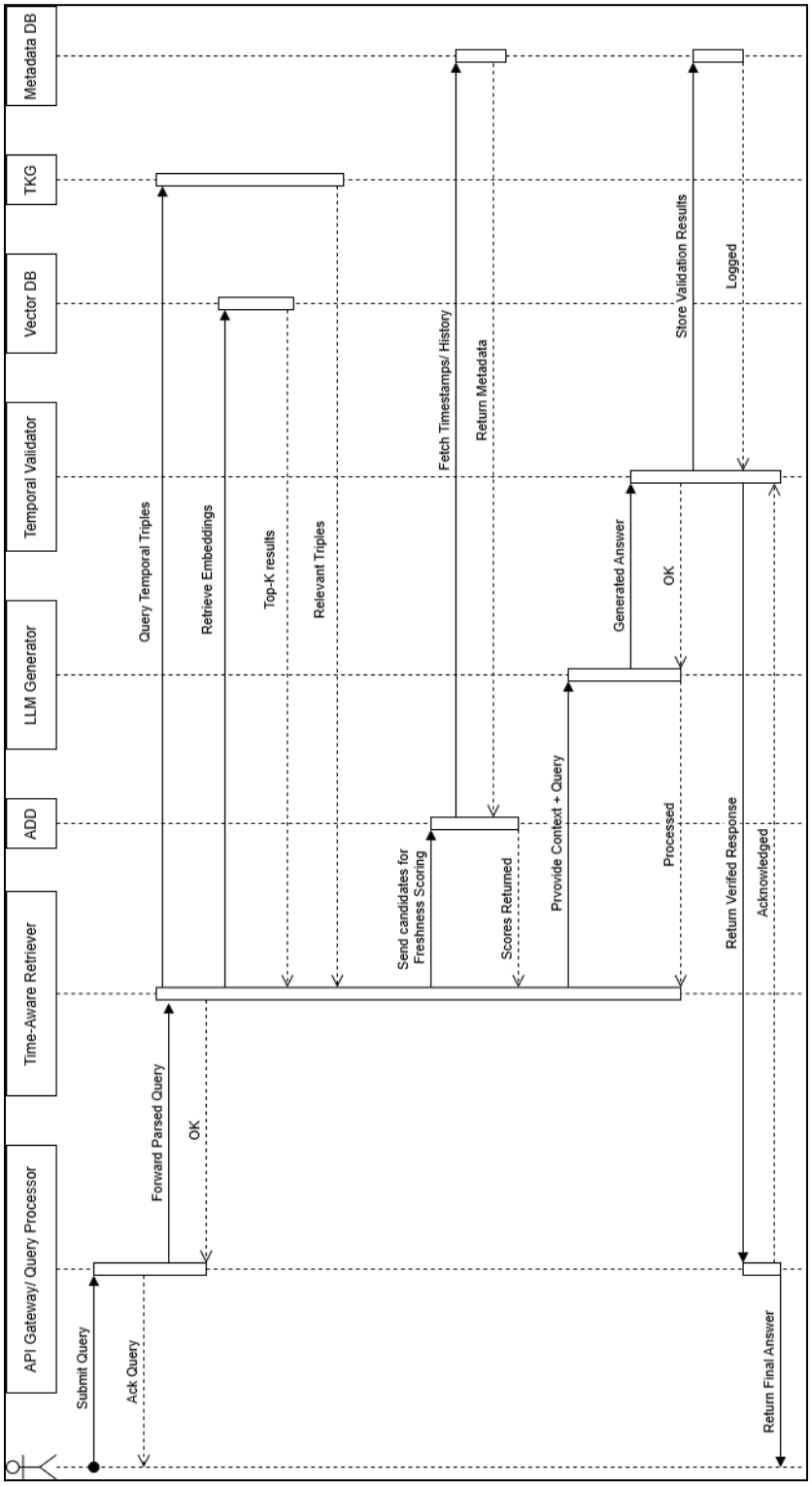

Fig. 6 - Sequence Diagram

METHODOLOGY AND PROPOSED APPROACH

4.1 Algorithms or Techniques to be Used

The proposed T-RAG framework binds several Artificial Intelligence and Knowledge Representation techniques that make time-aware reasoning possible, dynamic knowledge adaptation, and deprecation detection. The methodology is logically structured into three layers:

1. Knowledge Representation
2. Temporal Reasoning & Detection
3. Generative Integration.

(a) Temporal Knowledge Graph Construction (TKG Module)

Purpose: The aim is to express facts that are timestamped and relationships between entities evolving over time.

Technique Used:

- GNN-based embedding, like RGCN or RE-GCN, is used to model interactions between entities, relations, and time.
- Each fact is represented as a quadruple: head, relation, tail, and timestamp.
- Historical context is preserved by using GRU–based temporal updates.

Mathematical Formulation:

$$h_t = GRU(h_{t-1}, f(e_h, r, e_t))$$

where f(·) is the relational transformation, and ht is the entity embedding at time t.

Expected Output: A temporal knowledge graph embedding space can retrieve the facts which were valid at any given point in time.

(b) Adaptive Deprecation Detection (ADD Module)

The purpose would be to automatically identify and filter outdated or invalid facts using a learned temporal decay model.

Technique Used:

- Half-life-decay function inspired by HALO (2025) [3].
- A supervised ML model (Logistic Regression / Gradient Boosting) is trained to predict the FVS, based on historical usage, frequency of updates, and semantic drift.

Mathematical Model:

$$FVS = e^{-\lambda(t - t_{last})}$$

where λ is the decay rate learned from data and t_{last} is the last verification timestamp.

Algorithm (in simplified pseudocode):

```
For every fact f in TKG:
    Compute Δt = current_time - last_verified(f)
    Predict λ using the trained ML model
    FVS = exp(-λ * Δt)
    If FVS < threshold:
        Mark f as deprecated
```

Expected Output: The knowledge base used by the retriever, filtered and ranked by freshness.

(c) Time-Aware Retrieval-Augmented Generation (T-RAG)

Purpose: Combine semantically and temporally related facts for answer generation in LLM.

Technique Used:

- Hybrid Retrieval → This combines vector similarity search with temporal scoring.
- Weighted Relevance Score (WRS) computed as:

$$WRS = \alpha \cdot Sim(q, d) + (1 - \alpha) \cdot FVS$$

  where "Sim(q, d)" is the semantic similarity between query (q) and document (d), and (FVS) stands for temporal freshness [5].

- The top-k retrieved contexts are fed to the LLM (GPT/LLaMA) via a time-conditioned prompt.

Prompt Template Example:

Question: {user_query}

Context (valid as of {timestamp}):

These include the following: {filtered_context}

Please answer by considering only the most recent and valid information.

Expected Output: The response must be contextually correct, temporally aligned, and provide source timestamps for explainability.

(d) Evaluation and Validation

- **Metrics Used:**
  - Mean Reciprocal Rank (MRR) measures ranking quality.
  - Hits@k measures the correctness of the top-k retrieved results.
  - Temporal Accuracy (TA) measures the factual correctness with respect to the timestamp.
  - TCS (Temporal Consistency Score): A measure to what extent the model keeps timeline consistency.

- **Benchmarks:**
  - Datasets: ICEWS14, ICEWS05-15, and Wikidata Temporal.
  - Comparative models: RE-GCN, HALO, Chain-of-History, and LLM-DA [3], [5].

4.2 Description of Data Collection Process

(a) Data Sources

The project utilises open-source temporal knowledge datasets, representing event sequences from the real world with time annotations:

Dataset	Description	Time Granularity	Use in Project
ICEWS14	Integrated Crisis Early Warning System - political and social events	Daily	For testing the TKG and ADD module accuracy
ICEWS05-15	Extended event dataset for multi-year analysis	Daily	For assessing long-term decay and consistency in prediction.
Wikidata Temporal	Structured factual knowledge with start/end timestamps	Annual	For open-domain temporal reasoning and LLM integration

(b) Data Preprocessing

1. Entity and Relation Extraction: Named Entity Recognition (NER) and Dependency Parsing via spaCy/BERT NER.
2. Timestamp Normalisation: All time fields will be returned in ISO 8601 format.
3. Duplicate filtering: Remove repeated or conflicting records.
4. Embedding Generation: Obtain dense embeddings with Sentence-BERT or OpenAI Embeddings.

5. Storage: Store preprocessed facts in:
    - Neo4j for structured triples with timestamps.
    - FAISS / Milvus for semantic vector retrieval.

(c) Data Validation

- Use **temporal constraints** to guarantee logical consistency, such as

  $$\text{event_end} \geq \text{event_start}$$

- Use **random sampling** to manually inspect the correctness of the extracted timestamps and entities.
- Evaluate data quality by **freshness ratio**, i.e., the percentage of non-deprecated facts.

(d) Summary of Data Flow

Raw data ingested from ICEWS/Wikidata

↓

Parsed into entity–relation–time quadruples

↓

Cleaned and embedded

↓

Stored in Neo4j (TKG) + Vector DB

↓

Used by the Retriever and ADD modules during training and inference

EXPECTED OUTCOME

5.1 Expected Results

Functional Outcomes:

- Temporal Knowledge Graph for time-stamped reasoning.
- Adaptive Deprecation Detection module for filtering outdated knowledge.
- Time-Aware RAG pipeline for contextually and temporally accurate outputs.
- Explainable, timestamped LLM responses.

Quantitative Metrics:

Metric	Target	Improvement
Temporal Accuracy	≥90%	+20% vs baseline
Factual Consistency	≥85%	+25% over RAG
MRR	≥0.47	+0.03 gain
Latency	≤3s	30% faster than baseline

Qualitative Outcomes:

- Improved temporal reliability and explainability.
- Adaptation to evolving data streams.
- Applicability to domains like news verification, healthcare, and finance.

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